








Advanced Time Series Analysis and Forecasting

Chapter 5: Bayesian VAR, factor models and nowcasting



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Acronyms used in this material (1/2)

- **ADL-MIDAS**: Autoregressive Distributed Lag MIDAS
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **BGR**: Bańbura–Giannone–Reichlin (large Bayesian VARs, 2010)
- **BIC**: Bayesian Information Criterion
- **BVAR**: Bayesian Vector AutoRegression
- **CPI**: Consumer Price Index
- **DFM**: Dynamic Factor Model
- **DI**: Diffusion Index (forecast)
- **DI-AR**: Diffusion Index forecast with AutoRegressive lags
- **EM**: Expectation–Maximisation (algorithm)
- **EMPL**: employment (total private payrolls)
- **ESI**: Economic Sentiment Indicator (European Commission)
- **ESS**: Effective Sample Size
- **FAVAR**: Factor-Augmented Vector AutoRegression
- **FFR**: Federal Funds Rate
- **FRED**: Federal Reserve Economic Data
- **FRED-MD**: FRED Monthly Database for macroeconomic research (McCracken–Ng)
- **GDP**: Gross Domestic Product
- **GLP**: Giannone–Lenza–Primiceri (hierarchical prior selection, 2015)

Acronyms used in this material (2/2)

- **INS**: Institutul Național de Statistică (the National Institute of Statistics (Romania))
- **IP**: Industrial Production
- **LARGE**: the large system of BGR (all series observed over the sample)
- **LLM**: Large Language Model
- **MCMC**: Markov Chain Monte Carlo
- **MEDIUM**: the 20-variable system of BGR
- **MIDAS**: MIxed DAta Sampling (regression)
- **ML**: Machine Learning
- **MSE**: Mean Squared Error
- **MSFE**: Mean Squared Forecast Error
- **NBER**: National Bureau of Economic Research
- **NIW**: Normal–inverse-Wishart (prior)
- **NLS**: Nonlinear Least Squares
- **OLS**: Ordinary Least Squares
- **PCA**: Principal Component Analysis
- **RMSE**: Root Mean Squared Error
- **SE**: Standard Error
- **SMALL**: the 3-variable system of BGR (employment, CPI, federal funds rate)
- **TSA**: Time Series Analysis
- **US**: United States
- **VAR**: Vector AutoRegression

Today's question and route

- **Question**: how do we forecast and analyse an economy described by a hundred series, some of them published monthly and late, others quarterly?
  - an unrestricted VAR with 100 variables and 13 lags has more than 130 000 coefficients: the data alone cannot pin them down
- **Route** of the chapter
  - the curse of dimensionality; Bayesian inference at the depth we need (conjugacy, Gibbs sampling, MCMC diagnostics)
  - Minnesota and Normal–inverse-Wishart priors, dummy observations, shrinkage chosen by the marginal likelihood, large BVARs
  - factor models: principal components, the number of factors, diffusion indexes, FAVAR, dynamic factor models
  - mixed frequencies and nowcasting: bridge equations, MIDAS, the Kalman filter with a ragged edge, news; Romanian GDP
- We build on TSA, Chapter 6 (VAR), TSA, Chapter 10 (Kalman filter) and Chapter 3 (structural VAR); Seminar 5 comes before this lecture

Learning outcomes

- Derive conjugate posteriors, run a Gibbs sampler and judge its convergence with trace plots, effective sample size and $\widehat{R}$
- Write the Minnesota prior in natural conjugate form and with dummy observations, and explain each hyperparameter
- Choose the tightness of a large BVAR by the marginal likelihood or by the in-sample fit rule, and evaluate the forecasts out of sample
- Estimate approximate factor models by principal components, select the number of factors and build diffusion-index forecasts and a FAVAR
- Nowcast quarterly GDP from monthly data with bridge equations, MIDAS and a dynamic factor model, and decompose each revision into news

Reading, data and tools

- Backbone: Kilian and Lütkepohl (2017), Ch. 5 (Bayesian VAR) and Ch. 16 (large VARs, factors); Karlsson (2013); Stock and Watson (2016)
  - Bayesian background: Gelman et al. (2013); state space: Durbin and Koopman (2012); nowcasting survey: Bańbura, Giannone, Modugno and Reichlin (2013)
- Python Quantlets of this chapter: Quantlets/Ch_05
  - Minnesota and Normal–inverse-Wishart BVAR, marginal likelihood, PCA with EM, Bai–Ng, FAVAR, Kalman filter and news written in `numpy`
  - comparison packages: `statsmodels` (DynamicFactorMQ, EM estimation of the nowcasting model) and `bvar` of the Bank of England (GLP hyperparameters)
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source	Use
117 US monthly series in the FRED-MD format (output, labour, housing, orders, money, rates, prices), 1960:1–2026:7	FRED (St. Louis Fed), one CSV per series; FRED-MD transformation codes McCracken and Ng (2016)	BVAR, factors, FAVAR
NBER recession dates; US capacity utilisation, unemployment rate	FRED (USREC, CUMFNS, UNRATE)	charts, Gibbs
Romania: IP, retail trade, construction, unemployment, six confidence indicators (11 monthly series); euro-area IP	Eurostat (sts_inpr_m, sts_trtu_m, sts_copr_m, une_rt_m, ei_bssi_m_r2)	nowcasting
Romanian real GDP, quarterly, seasonally and calendar adjusted	Eurostat (namq_10_gdp)	target

- All sources are public and need no account or key; FRED-MD codes: 1 level, 2 difference, 4 log, 5 log difference, 6 second log difference
- October 2025: the US household survey and CPI were not collected (federal shutdown); the isolated missing month is interpolated

Origins of Bayesian VARs: the Minneapolis Fed



Federal Reserve Bank of Minneapolis, 2010

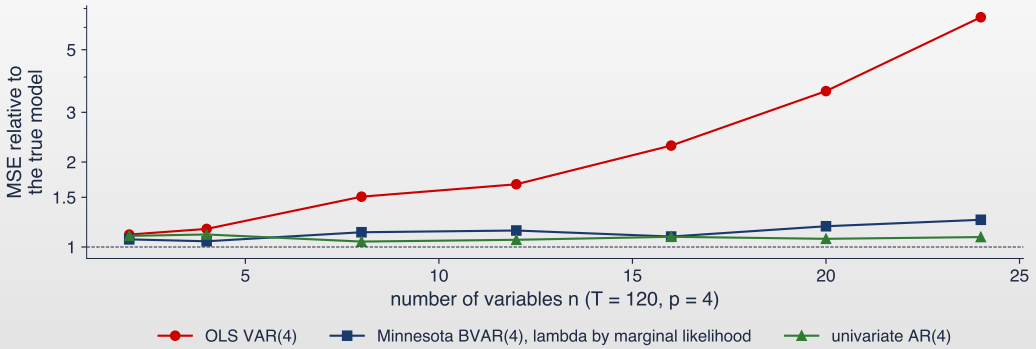
Photo: Innotata (2010); CC BY-SA 3.0; Wikimedia Commons

- Early 1980s: Litterman forecasts with Bayesian VARs at the Minneapolis Fed Litterman (1986); the "Minnesota prior"
  - Doan, Litterman and Sims (1984): sum-of-coefficients and conditional projections; Sims (1993), Sims and Zha (1998): dummy initial observations
- 1989–2002: dynamic factor indexes Stock and Watson (1989), generalised dynamic factors Forni, Hallin, Lippi and Reichlin (2000), diffusion indexes Stock and Watson (2002a)
- 2005–2015: FAVAR Bernanke, Boivin and Eliasz (2005), nowcasting Giannone, Reichlin and Small (2008), large BVARs Bańbura, Giannone and Reichlin (2010), hierarchical priors Giannone, Lenza and Primiceri (2015)
- One idea throughout: when parameters outnumber observations, add information (a prior) or reduce dimension (factors)

Counting parameters

- VAR(p) in n variables: $k = np + 1$ regressors per equation, nk slope coefficients and $n(n+1)/2$ covariances
  - $n = 3$, $p = 13$: $k = 40$; $n = 20$: $k = 261$; $n = 102$: $k = 1\,327$ per equation
  - a 10-year window has $T = 120$ months: with $k > T$, OLS does not exist ($X'X$ is singular)
- Even with $k < T$, the estimation error grows with k/T: one-step MSE $\approx \sigma^2(1 + k/T)$ for a correctly specified regression
  - overfitting: in-sample fit improves, out-of-sample accuracy deteriorates
- Three remedies
  - shrinkage (Bayesian priors, ridge, LASSO): a small bias for a large reduction of variance
  - dimension reduction by factors: a few common components summarise many series
  - variable selection: keep only some regressors

Simulation: OLS VAR against a Minnesota BVAR



- True model: stationary VAR(1) $A = 0.5I + (0.3/n)11'$ (1: a vector of ones); estimated VAR(4), $T = 120$; one-step MSE of variable 1 relative to the true model; 200 replications

ATS_ch5_bvar

Interpreting the simulation

- OLS: the relative MSE climbs to 6.52 at $n = 24$ ($k = 97$ regressors for 116 observations)
- BVAR: 1.25 at $n = 24$; the tightness chosen by the marginal likelihood falls from 0.31 ($n = 2$) to 0.14 ($n = 24$)
- A univariate AR(4) stays near 1.08: the cross-variable information is real but small, and OLS spends it on noise
- The larger the system, the tighter the prior should be: the theme of the whole chapter De Mol, Giannone and Reichlin (2008)

Prior, likelihood, posterior



Pierre-Simon Laplace, who developed inverse probability

Portrait: Paulin Guérin (1838); public domain; Wikimedia Commons

- Bayes: $p(\theta \mid y) = p(y \mid \theta)p(\theta)/p(y)$, with $p(y) = \int p(y \mid \theta)p(\theta)\, d\theta$
  - θ: the parameters; y: the data; $p(\theta)$: **prior**; $p(y \mid \theta)$: likelihood; $p(\theta \mid y)$: **posterior**
  - $p(y)$: **marginal likelihood**, the evidence for the model and its prior
- Conjugacy: prior and posterior in the same family, so the posterior has a closed form
  - Normal mean with known variance: Normal; regression coefficients and variance: Normal–inverse-gamma
- Point summaries: posterior mean (quadratic loss), median (absolute loss); credible sets from posterior quantiles

The Normal update is a precision-weighted average (1/2)

- Data: $\hat{\beta} \mid \beta \sim N(\beta, s^2)$ with $s^2 = \sigma^2 / \sum x_t^2$; prior: $\beta \sim N(m_0, v_0)$
  - $\hat{\beta}$: the OLS estimate of a slope β on regressor x_t; σ^2: the error variance, taken as known; s^2: the sampling variance of $\hat{\beta}$
  - m_0, v_0: the prior mean and variance; precision $= 1/\text{variance}$
- Complete the square in β: the posterior is Normal, $\beta \mid y \sim N(m_1, v_1)$
  - precisions add: $v_1^{-1} = v_0^{-1} + s^{-2}$
  - means average: $m_1 = w\, m_0 + (1 - w)\hat{\beta}$, with prior weight $w = v_0^{-1} / (v_0^{-1} + s^{-2}) \in (0, 1)$
  - w is large when the prior is tight (v_0 small) or the data are weak (s^2 large)

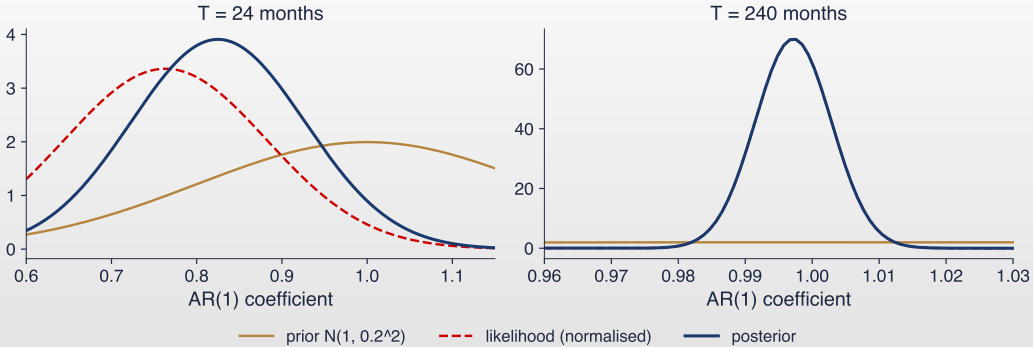
The Normal update is a precision-weighted average (2/2)

- In regression form:

$$\bar{\beta} = (V_0^{-1} + X'X/\sigma^2)^{-1}(V_0^{-1}m_0 + X'y/\sigma^2)$$

  - y ($T \times 1$), X ($T \times k$): the data; m_0, V_0: prior mean vector and covariance matrix; $\bar{\beta}$: the posterior mean
  - the same weighting: prior precision V_0^{-1} plus data precision $X'X/\sigma^2$
- With $m_0 = 0$ and $V_0 = \tau^2 I$ this is ridge regression with penalty σ^2/τ^2
  - τ^2: the prior variance of every coefficient; a small τ^2 shrinks more towards 0
- As $T \to \infty$, $X'X$ grows and the prior weight $w \to 0$: the prior matters in small samples and in large models
- Seminar 5, A1: the update by hand and the empirical-Bayes choice of τ^2

Updating an AR(1) coefficient



- US unemployment rate, monthly, windows ending December 2019; prior $N(1, 0.2^2)$ (random walk), σ^2 at its OLS estimate

ATS_ch5_bayes

Interpreting the update

- $T = 24$: OLS 0.763 (SE 0.119), posterior mean 0.825 (sd 0.102); the prior has weight 26.0%
- $T = 240$: OLS 0.997, posterior 0.997; the prior weight falls to 0.1%
- In a VAR, each equation has hundreds of coefficients but the same T: the prior keeps the weight it has at $T = 24$ here

Marginal likelihood and Bayes factors

- $p(y \mid M) = \int p(y \mid \theta, M) p(\theta \mid M)\, d\theta$: the density of the data *before* seeing them
  - M: a model, i.e. a likelihood together with its prior; the integral averages the likelihood over the prior
  - prediction-error decomposition: $\ln p(y) = \sum_t \ln p(y_t \mid y_{1:t-1})$, a sum of one-step log scores (Chapter 1); $y_{1:t-1}$: the data up to $t-1$
  - it penalises complexity automatically: a diffuse prior spreads probability over data that never occur
- Bayes factor $B_{12} = p(y \mid M_1)/p(y \mid M_2)$; $2 \ln B_{12} > 6$ is "strong" evidence Kass and Raftery (1995)
  - sensitive to the prior: with an improper prior it is not defined
- Hierarchical use: treat the hyperparameters γ of the prior as parameters and maximise $p(y \mid \gamma)$ (empirical Bayes) or put a hyperprior on them Giannone, Lenza and Primiceri (2015)

Gibbs sampling (1/2)

- Use: the joint posterior has no closed form, but each **full conditional** has one Geman and Geman (1984), Gelfand and Smith (1990)
  - draw $\theta_1 \mid \theta_2, y$, then $\theta_2 \mid \theta_1, y$, and repeat
  - θ_1, θ_2: two blocks of parameters; full conditional: the distribution of one block given the other and the data
  - the draws form a Markov chain whose stationary distribution is the posterior
  - discard a burn-in, then average functions of the draws (ergodic theorem)
- In VARs: stochastic volatility, non-conjugate priors and set-identified SVARs (Chapter 3) need Gibbs or Metropolis–Hastings steps

Gibbs sampling (2/2): the linear regression

- Regression $y = X\beta + e$ with independent priors $\beta \sim N(b_0, V_0)$, $\sigma^2 \sim IG(a_0, d_0)$
  - b_0, V_0: prior mean and covariance of β; $IG(a_0, d_0)$: inverse gamma with shape a_0 and scale d_0
- Step 1, the coefficients given the variance: $\beta \mid \sigma^2, y \sim N(\bar{b}, \bar{V})$
  - $\bar{V} = (V_0^{-1} + X'X/\sigma^2)^{-1}$, $\bar{b} = \bar{V}(V_0^{-1} b_0 + X'y/\sigma^2)$: the Normal update of the previous slides
- Step 2, the variance given the coefficients: $\sigma^2 \mid \beta, y \sim IG(a_0 + T/2,\ d_0 + e'e/2)$
  - $e = y - X\beta$: the residuals at the current draw of β; the data add $T/2$ to the shape and half the residual sum of squares to the scale

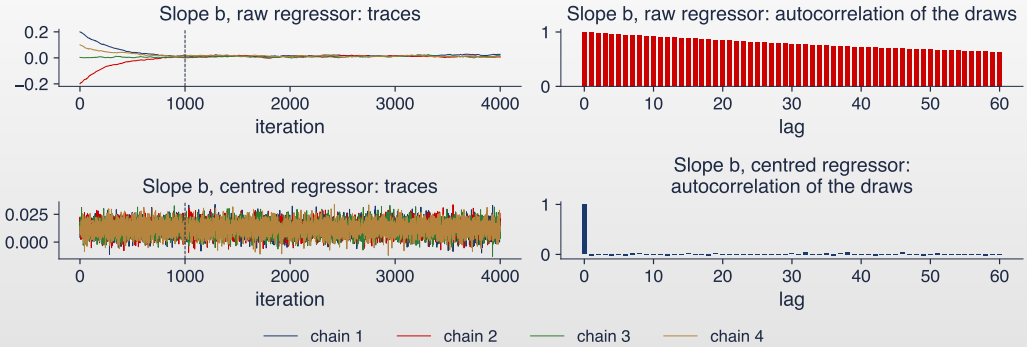
MCMC diagnostics (1/2)

- **Effective sample size**: $\mathrm{ESS} = M/(1 + 2\sum_{k\geq 1}\rho_k)$: the number of independent draws with the same precision
  - M: the number of draws kept; ρ_k: the autocorrelation of the draws at lag k
  - AR(1)-type chain: $\mathrm{ESS} = M(1-\rho)/(1+\rho)$; $\rho = 0.9$ keeps about 5% of the draws
- $\widehat{R}$ Gelman and Rubin (1992): several chains from dispersed starts; $\widehat{R} = \sqrt{\hat{V}/W}$, $\hat{V} = \frac{n-1}{n}W + B/n$
  - n: draws per chain; W: within-chain variance; B: $n\times$ the variance of the chain means; $\hat{V}$: the pooled variance estimate
  - $\widehat{R} \approx 1$: the chains agree

MCMC diagnostics (2/2)

- $\widehat{R}$ in current practice Vehtari et al. (2021)
  - rank-normalised, split chains; require $\widehat{R} < 1.01$
- **Geweke** z: $z = (\bar{\theta}_A - \bar{\theta}_B)/\sqrt{\hat{S}_A(0)/n_A + \hat{S}_B(0)/n_B}$
  - $\bar{\theta}_A$, $\bar{\theta}_B$: the means of the first 10% and of the last 50% of a chain; n_A, n_B: their lengths
  - $\hat{S}(0)$: the spectral density at frequency zero, which accounts for autocorrelation
  - $z \approx N(0,1)$ after convergence; $|z| > 2$ signals non-convergence
- None of the three proves convergence; each can reveal a failure

Gibbs sampling: a bad and a good parametrisation



- Monthly US IP growth on lagged capacity utilisation (mean 79.0%, sd 4.7), 1967–2019, $T = 635$; one-at-a-time Gibbs, four chains, 4 000 draws, burn-in 1 000

ATS_ch5_bayes

Interpreting the two samplers

- Raw regressor: the posterior correlation of intercept and slope is -0.998, so each conditional step moves very little
  - lag-1 autocorrelation 0.992; ESS $= 53$ out of 12 000 draws; $\widehat{R} = 1.32$; Geweke $z = -2.27$
- Centred regressor: correlation -0.001, ESS $= 11\,529$, $\widehat{R} = 1.00$, Geweke $z = 0.01$
  - same model, same posterior of the slope (mean 0.0123); only the sampler changed
- Lesson: reparametrise or draw correlated blocks jointly; conjugate BVARs avoid the problem by drawing B in one block

The Minnesota prior

- VAR(p): $y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t$; $(A_l)_{ij}$: the effect of variable j at lag l on variable i
  - prior means Litterman (1986): $\mathbb{E}[(A_1)_{ii}] = \delta_i$, all other coefficients 0; $\delta_i = 1$ (random walk) for persistent series, 0 (white noise) otherwise
  - prior variances Bańbura, Giannone and Reichlin (2010), eq. (2): $\text{Var}[(A_l)_{ij}] = \lambda^2 / l^2$ if $j = i$, $\vartheta \lambda^2 \sigma_i^2 / (l^2 \sigma_j^2)$ if $j \neq i$
- Hyperparameters
  - λ: overall tightness; $1/l^2$: the decay with the lag; $\vartheta \in (0, 1]$: cross-variable shrinkage
  - σ_i^2: residual variance of a univariate AR for y_i, so that σ_i^2 / σ_j^2 fixes units
  - $\lambda = 0$: posterior = prior; $\lambda \to \infty$: posterior mean = OLS
- Original version: Σ diagonal and fixed, each equation a separate ridge-type regression with a diffuse constant

The natural conjugate prior (1/2)

- The VAR as a multivariate regression $Y = XB + U$, rows of U i.i.d. $N(0, \Sigma)$
  - Y ($T \times n$): the observations; X ($T \times k$): constant and p lags; B ($k \times n$): the coefficients, one column per equation
- Prior Kadiyala and Karlsson (1997): $\mathrm{vec}(B) \mid \Sigma \sim N(\mathrm{vec}(B_0), \Sigma \otimes \Omega_0)$, $\Sigma \sim IW(\Psi, d)$
  - vec: stacks the columns; $\otimes$: Kronecker product; B_0: prior mean; Ω_0 ($k \times k$): prior covariance across regressors
  - $IW(\Psi, d)$: inverse Wishart with scale matrix Ψ and d degrees of freedom
- Posterior: the same families, with updated parameters (derivation: ▸ Appendix)
  - $\bar{\Omega}^{-1} = \Omega_0^{-1} + X'X$, $\bar{B} = \bar{\Omega}(\Omega_0^{-1} B_0 + X'Y)$: the precision-weighted average again
  - $\Sigma \mid Y \sim IW(\bar{\Psi}, T + d)$, $\bar{\Psi} = \Psi + \hat{U}'\hat{U} + (\bar{B} - B_0)'\Omega_0^{-1}(\bar{B} - B_0)$, $\hat{U} = Y - X\bar{B}$

The natural conjugate prior (2/2)

- Price of conjugacy: the Kronecker structure gives every equation the same Ω_0, so $\vartheta = 1$
  - Minnesota moments with $\Omega_0 = \mathrm{diag}(\lambda^2/(l^2\sigma_j^2))$ and $\mathbb{E}\Sigma = \mathrm{diag}(\sigma_i^2)$ ($d = n + 2$, $\Psi = \mathrm{diag}(\sigma_i^2)$)
- Gain: one $k \times k$ inversion for all equations; exact draws without MCMC; a closed-form marginal likelihood (▸ Appendix)

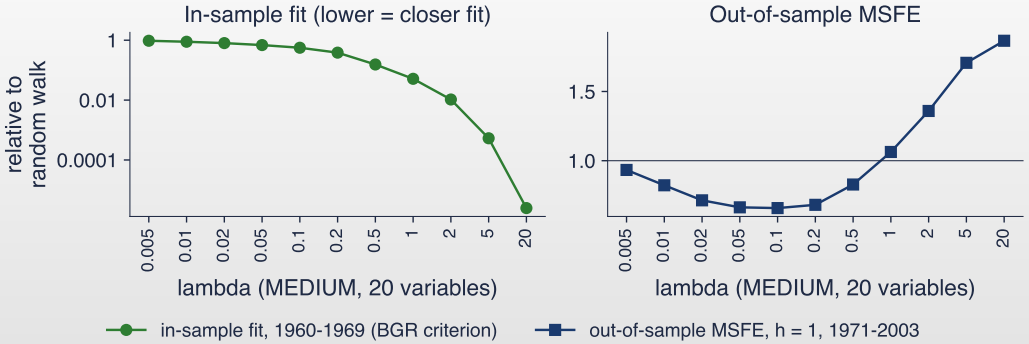
Dummy observations (1/2)

- Theil mixed estimation
  - add T_d artificial rows (Y_d, X_d) and run OLS on $\binom{Y_d}{Y}$, $\binom{X_d}{X}$
  - each row states a prior belief as if it were data
  - equivalent to the NIW prior with $B_0 = (X_d'X_d)^{-1}X_d'Y_d$, $\Omega_0 = (X_d'X_d)^{-1}$ Bańbura, Giannone and Reichlin (2010), eq. (5)
  - Minnesota block: $Y_d = \mathrm{diag}(\delta_i\sigma_i)/\lambda$, $X_d = J_p \otimes \mathrm{diag}(\sigma_i)/\lambda$, $J_p = \mathrm{diag}(1, \ldots, p)$

Dummy observations (2/2)

- **Sum of coefficients** Doan, Litterman and Sims (1984): $Y_d = \text{diag}(\bar{y}_i)/\mu$, $X_d = (1'_p \otimes \text{diag}(\bar{y}_i)/\mu,\ 0)$
  - $\bar{y}_i$: the mean of the first p observations of y_i; 1_p: a vector of p ones; $\mu > 0$: the tightness of this prior
  - shrinks $\Pi = I - \sum_l A_l$ to 0 ("inexact differencing"): $\mu \to 0$ gives a VAR in differences without cointegration
- **Dummy initial observation** Sims (1993), Sims and Zha (1998): one row $y = \bar{y}'/\phi$, $x = (\bar{y}'/\phi, \ldots, \bar{y}'/\phi, 1/\phi)$
  - $\bar{y}$: the vector of the $\bar{y}_i$; $\phi > 0$: its tightness
  - pushes towards unit roots *or* towards a stationary model whose mean is $\bar{y}$
  - so it allows cointegration (Chapter 4)

How much shrinkage? In-sample fit against out-of-sample accuracy



- MEDIUM system (20 variables, $p = 13$, $k = 261$), BGR prior, rolling 10-year windows; out-of-sample: one-step MSFE of employment, CPI and the funds rate, 1971–2003, relative to a random walk

ATS_ch5_large_bvar

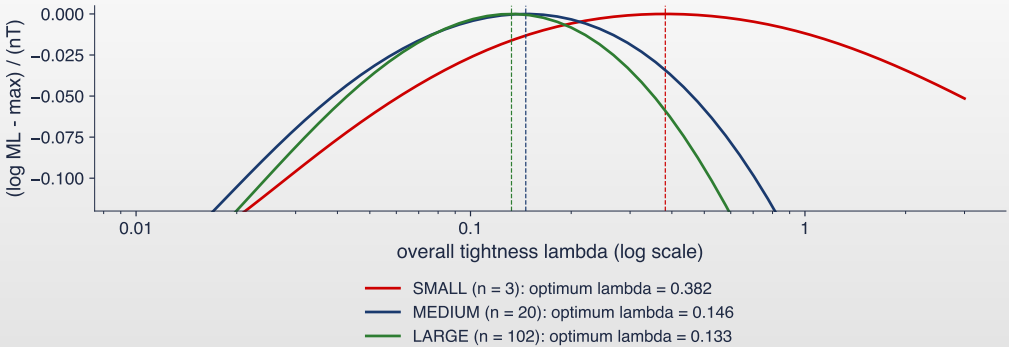
Interpreting the trade-off

- The in-sample fit improves without limit as λ grows: with $k = 261$ and $T = 120$, a loose prior interpolates the data
- Out of sample: U-shape; minimum 0.66 at $\lambda = 0.10$; 0.93 with $\lambda = 0.005$ (almost the random walk); 1.87 with $\lambda = 20$
- The shrinkage must be chosen without looking at the evaluation sample: by a fit rule (BGR) or by the marginal likelihood (GLP)

Giannone, Lenza and Primiceri (2015)

- Treat $\gamma = (\lambda, \mu, \phi)$ as parameters with hyperpriors: $p(\gamma \mid y) \propto p(y \mid \gamma)p(\gamma)$ Giannone, Lenza and Primiceri (2015)
  - $p(y \mid \gamma)$ is known in closed form for the NIW prior with dummies: data plus dummies, minus dummies alone
  - hyperpriors: Gamma with mode 0.2 and sd 0.4 for λ, mode 1 and sd 1 for μ and ϕ
- Interpretation: the marginal likelihood is an out-of-sample score of one-step predictive densities (slide Marginal likelihood and Bayes factors)
  - so it trades fit against complexity without an evaluation sample
- Use the posterior mode of γ (as here) or integrate γ out by Metropolis–Hastings; GLP find both forecast about as well as factor models

The marginal likelihood as a function of the tightness



- SMALL ($n = 3$), MEDIUM ($n = 20$), LARGE ($n = 102$), $p = 13$, 1960:1–2019:12 ($T = 707$); $\mu = \phi = 1$; log ML minus its maximum, divided by nT

ATS_ch5_large_bvar

Interpreting the marginal likelihood

- Optimal tightness: 0.382 (SMALL), 0.146 (MEDIUM), 0.133 (LARGE): bigger systems need a tighter prior
- Using the textbook value $\lambda = 0.2$ costs -46.9 log points for MEDIUM and -597.2 for LARGE; $\lambda = 1$ costs -15571.6 for LARGE
- Joint mode with the hyperpriors: SMALL $\mu = 0.20$, $\phi = 0.44$; MEDIUM $\mu = 0.13$, $\phi = 0.44$: the data ask for tight sum-of-coefficients and initial-observation priors

Choosing the overall tightness: two rules (1/2)

- **BGR fit rule** Bańbura, Giannone and Reichlin (2010), Section 3
  - choose λ so that the in-sample one-step fit of the key variables equals that of a small OLS VAR
  - $\mathrm{Fit}(\lambda) = \frac{1}{3}\sum_{i \in I} \mathrm{msfe}_i^{(\lambda)} / \mathrm{msfe}_i^{(0)}$ on 1960–1969; here the target is 0.43
  - I: the three key variables; $\mathrm{msfe}_i^{(\lambda)}$: in-sample one-step mean squared error of variable i with tightness λ; $\lambda = 0$: the prior mean alone
  - our values: MEDIUM 0.158, LARGE 0.059 (BGR Table 1: 0.108 and 0.035 on their data set)

Choosing the overall tightness: two rules (2/2)

- **GLP marginal likelihood** Giannone, Lenza and Primiceri (2015)
  - the mode of λ is re-estimated on each window
  - on 10-year windows the median mode is 0.34 (SMALL) and 0.28 (MEDIUM)
  - 10%–90% range of the MEDIUM mode across windows: [0.23, 0.33]
- The Python package `bvar` of the Bank of England implements the GLP optimisation for the conjugate model; the notebook compares it with our `numpy` code

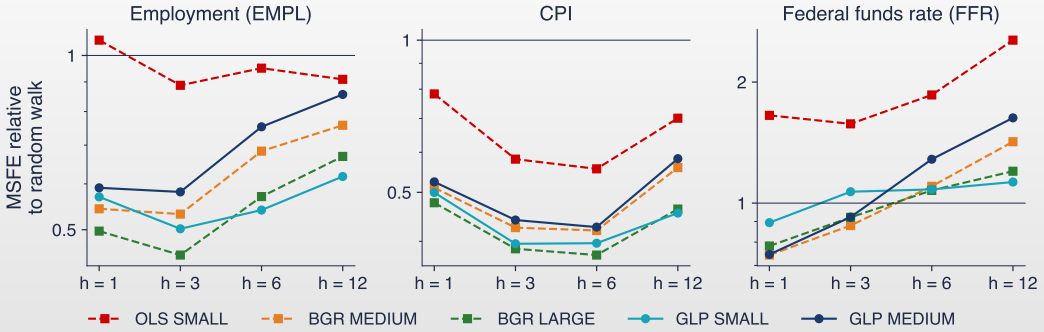
Bańbura, Giannone and Reichlin (2010): the design

- Bańbura, Giannone and Reichlin (2010)
  - question: can a VAR with 131 variables forecast and identify shocks?
  - systems: SMALL (employment, CPI, funds rate), MEDIUM (20), LARGE (all)
  - $p = 13$; rolling 10-year windows; posterior-mean point forecasts iterated to $h = 12$; benchmark: random walk with drift
  - Minnesota NIW prior plus sum-of-coefficients with $\tau = 10\lambda$ (Section 3.3); λ by the fit rule on 1960–1969

Bańbura, Giannone and Reichlin (2010): our replication

- Our data: FRED-MD-format panel; LARGE = the 102 series observed over 1960–2026 (aggregates and components)
  - MEDIUM: the BGR list, with three substitutions
  - the S&P 500, the effective exchange rate and nonborrowed reserves are replaced by the Baa yield, the 3-month bill rate and building permits
  - reason: not available for 1960–2026, or negative after 2008
- Evaluation
  - targets 1971–2003 (the paper), 2004–2019 and 2004–2026 (extensions)
  - 678 forecast origins; GLP SMALL and MEDIUM added

Forecast accuracy, 1971–2003



⊡ MSFE relative to the random walk with drift (log scale; below 1 = better), employment and CPI in 100 × log levels, funds rate in percent

ATS_ch5_large_bvar

Interpreting the forecast comparison

- $h = 1$, employment: OLS SMALL 1.06, BGR MEDIUM 0.54, LARGE 0.50 (BGR Table 1: 1.14, 0.54, 0.46)
  - CPI: 0.78, 0.51, 0.48; funds rate: 1.65, 0.74, 0.78
- Adding variables with more shrinkage helps: LARGE beats SMALL for every variable at $h \leq 6$
- GLP SMALL is nearly as good as the large systems for employment and CPI (0.57, 0.50): most of the gain comes from the prior, not only from the variables
- The funds rate beyond 3 months is hard for all models (LARGE 1.20 at $h = 12$)

After 2003: the extension and March 2020

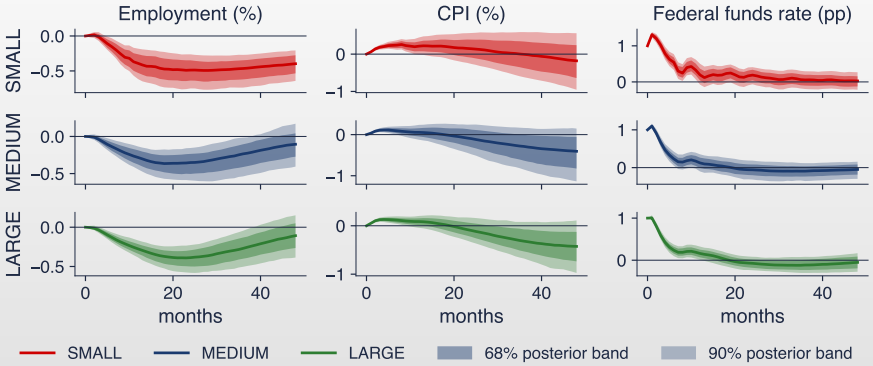
Model	EMPL, 1	EMPL, 12	CPI, 1	CPI, 12	FFR, 1	FFR, 12
OLS SMALL	0.38	1.08	1.26	2.43	0.98	3.51
BGR MEDIUM	0.20	0.66	0.97	6.49	0.55	1.91
BGR LARGE	0.19	0.67	0.92	3.73	0.45	1.79
GLP SMALL	0.24	0.33	0.93	1.07	0.61	0.67
GLP MEDIUM	0.20	0.28	0.95	2.55	0.57	0.91

- Targets 2004:1–2019:12, relative MSFE (random walk with drift = 1)
- Including 2020–2026, the 12-month employment MSFE explodes: BGR MEDIUM 500.06, GLP SMALL 8.64: windows containing April 2020 fit the collapse as dynamics
- Remedy: scale the residual variance of the pandemic months by estimated factors Lenza and Primiceri (2022), or stochastic volatility (next section)

A monetary policy shock in a large VAR

- Recursive scheme of Bańbura, Giannone and Reichlin (2010), Section 4: $y_t = (X_t, r_t, Z_t)$, slow variables X_t (real activity, prices), the funds rate r_t, fast variables Z_t (money, rates, exchange rates)
  - only the position of r_t matters (Chapter 3): no ordering inside the blocks is needed
- Bayesian bands: for each draw (B, Σ) from the NIW posterior, compute the Cholesky impact column and the responses; report posterior quantiles
  - 500 draws, 100 bp shock, sample 1961–2002, $p = 13$; SMALL with a nearly flat prior
- These are the posterior bands Chapter 3 announced: pointwise credible intervals, conditional on the identifying assumption

Responses to a 100 bp funds-rate shock



- Posterior medians with 68% and 90% bands; rows: SMALL, MEDIUM, LARGE; employment and CPI in percent

ATS_ch5_large_bvar

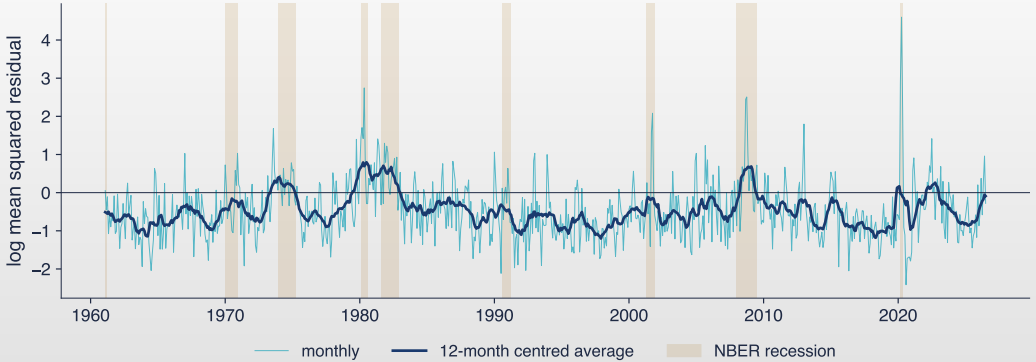
Interpreting the responses

- Employment: trough -0.49% (SMALL, month 28), -0.39% (LARGE, month 23); after four years LARGE is back to -0.11% (band $[-0.39, 0.15]$)
- CPI: the price puzzle shrinks from 0.26% (SMALL) to 0.13% (LARGE); after four years LARGE gives -0.43% (band $[-0.97, 0.11]$)
- As in BGR: more information makes the employment response less persistent and the price response more plausible
- The bands do not widen with 102 variables: the prior pays for the extra coefficients

Large BVARs with stochastic volatility

- **Common volatility** Carriero, Clark and Marcellino (2016): $u_t = \sqrt{f_t}\,\varepsilon_t$, $\varepsilon_t \sim N(0, \Sigma)$, $\ln f_t = \phi \ln f_{t-1} + \nu_t$
  - u_t: the VAR residual; $f_t > 0$: a volatility factor common to all equations; ϕ: its persistence; ν_t: its shock
  - given $f_{1:T}$, the model is a weighted conjugate BVAR (rows divided by $\sqrt{f_t}$); f_t is drawn by a Kim–Shephard–Chib step
- **Variable-specific volatilities** Carriero, Clark and Marcellino (2019), Clark (2011): $\Sigma_t = A^{-1} H_t A^{-1\prime}$, H_t diagonal with log-AR volatilities Primiceri (2005)
  - A: lower triangular with unit diagonal (the contemporaneous relations); H_t: the time-varying variances of the orthogonalised shocks
  - conjugacy is lost; the triangular algorithm draws the equations one at a time and makes $n = 100$ feasible
- Gains: mainly in density forecasts (calibrated intervals) and in robustness to outliers such as 2020

A common volatility factor in the BVAR residuals



⊡ MEDIUM BVAR (GLP hyperparameters), 1960–2026: log of the cross-sectional mean of the squared standardised residuals

ATS_ch5_large_bvar

Interpreting the volatility proxy

- Common movements: the residual variance rises in every recession and falls in the Great Moderation (1960–1984 is 1.28 times 1985–2007)
- April 2020 is 176 times the median month: a homoskedastic Gaussian BVAR treats it as an ordinary observation
- This is the case for the common-volatility model (one extra state) before the full variable-specific model; Chapter 6 gives the filtering tools

The approximate factor model

- $x_{it} = \lambda_i' F_t + e_{it}$, $i = 1, \ldots, N$, $t = 1, \ldots, T$; F_t: r common factors, λ_i: loadings, e_{it}: idiosyncratic
  - *approximate* Chamberlain and Rothschild (1983): the e_{it} may be weakly correlated across i and over t, provided the correlation is bounded as $N \to \infty$
  - Λ ($N \times r$): the stacked loadings; X ($T \times N$): the panel; Σ_Λ: a positive definite limit
  - pervasiveness: $\Lambda'\Lambda/N \to \Sigma_\Lambda > 0$, every factor affects many series; the r largest eigenvalues of XX' grow with N, the others do not
- Identification: $\Lambda F_t = (\Lambda H)(H^{-1} F_t)$ for any invertible H: only the factor space is identified
  - normalisation for PCA: $F'F/T = I_r$ and $\Lambda'\Lambda$ diagonal
- Static form of a dynamic model: lags of the dynamic factors are stacked in F_t Stock and Watson (2016); the generalised dynamic model works in the frequency domain Forni, Hallin, Lippi and Reichlin (2000)

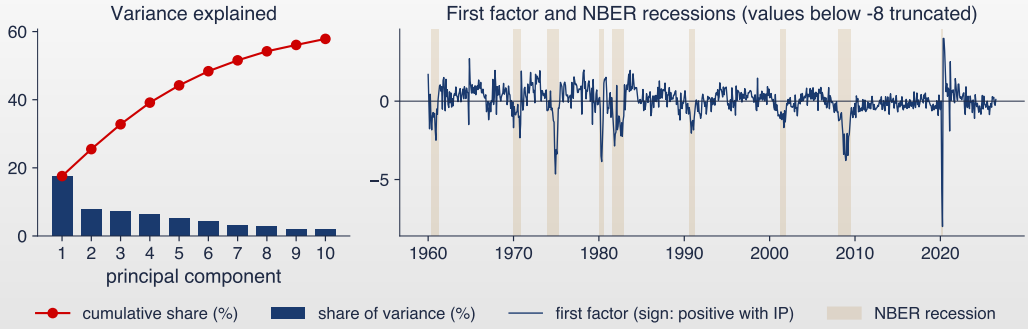
Principal components

- Least squares: $\min_{F,\Lambda} \sum_{i,t}(x_{it} - \lambda_i' F_t)^2$ with $F'F/T = I$: $\hat{F} = \sqrt{T} \times$ the first r eigenvectors of XX', $\hat{\Lambda} = X'\hat{F}/T$
  - standardise each series first: PCA is not scale invariant
- Consistency Stock and Watson (2002a), Bai (2003): $\hat{F}_t \to HF_t$ as $N, T \to \infty$ at rate $\min(\sqrt{N}, \sqrt{T})$
  - H: an invertible $r \times r$ rotation; PCA recovers the factors up to this rotation
  - if $\sqrt{T}/N \to 0$, the estimated factors can be used as regressors as if observed (no generated-regressor correction)
- Missing values and mixed start dates: EM Stock and Watson (2002b), McCracken and Ng (2016): fill with 0, run PCA, refill the missing cells with $\hat{\lambda}_i' \hat{F}_t$, iterate

The FRED-MD database

- McCracken and Ng (2016): about 130 US monthly series since 1959, updated monthly, with a transformation code per series and vintages for real-time work
  - eight groups: output and income, labour market, housing, consumption and orders, money and credit, interest and exchange rates, prices, stock market
  - outliers: an observation more than 10 interquartile ranges from the median is set to missing (3 cells here)
- Our panel: 117 FRED-MD series that FRED distributes (stock-market series are not), 1960:1–2026:7, 4.5% missing cells
  - the official file (current.csv of the St. Louis Fed) can replace it in the notebook with one line
- Current vintage only: the results are pseudo out of sample, not real time Croushore and Stark (2001)

Factors of the FRED-MD panel



- Left: share of the variance of the standardised panel explained by each principal component; right: the first factor (EM, 8 factors), NBER recessions shaded

ATS_ch5_factors

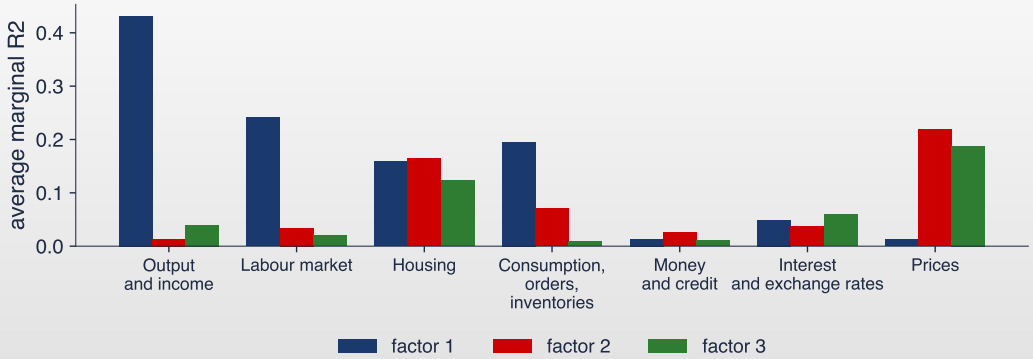
Interpreting the factors

- The first component explains 17.6%, the second 7.9%; five explain 44.2%, eight 54.2%
- The first factor is a real activity index: correlation -0.58 with the NBER recession indicator; its minimum is April 2020
- No component dominates: macro panels have a few strong factors and a long tail of weak ones, which makes the number of factors a real question

How many factors? Bai and Ng (2002)

- $V(k) = (NT)^{-1}\sum_{i,t}(x_{it} - \hat{\lambda}_i'\hat{F}_t)^2$ after k components; it falls with k, so it needs a penalty Bai and Ng (2002)
  - $IC_{p1}(k) = \ln V(k) + k\frac{N+T}{NT}\ln\frac{NT}{N+T}$, $IC_{p2}(k) = \ln V(k) + k\frac{N+T}{NT}\ln C_{NT}^2$, $C_{NT}^2 = \min(N, T)$
  - $IC_{p3}(k) = \ln V(k) + k\ln C_{NT}^2/C_{NT}^2$; choose the k that minimises the criterion; the penalty grows with k and vanishes as $N, T \to \infty$
  - PC_p versions: $V(k) + k\hat{\sigma}^2 g(N, T)$ with $\hat{\sigma}^2 = V(k_{\max})$, $g(N, T)$ the same penalties: depend on $k_{\max}$, the largest k tried
- Our panel ($k_{\max} = 15$): $IC_{p1} = 8$, $IC_{p2} = 8$, $IC_{p3} = 10$, $PC_{p1} = 13$, $PC_{p2} = 11$
  - the criteria disagree when the eigenvalues decline slowly; report the choice and its sensitivity
- For forecasting, the number of factors in the regression is chosen separately (by BIC), as in diffusion-index forecasts

The economic content of the factors



- Average marginal R^2 of the regression of each standardised series on one factor, by group (as in McCracken and Ng 2016)

ATS_ch5_factors

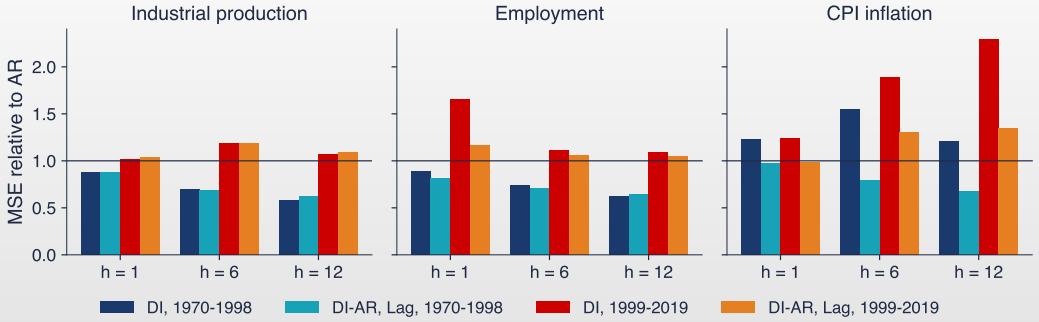
Interpreting the marginal R^2

- Factor 1 is real activity: R^2 0.43 for output, 0.24 for labour; the best single series has 0.78
- Factors 2 and 3 load on prices (0.22, 0.19) and housing (0.16, 0.12)
- Interest rates load weakly on the first three factors: monetary information sits in later factors or in the rates themselves, the motivation for the FAVAR
- Labels are a rotation choice: the factor space, not each factor, is identified

Diffusion-index forecasts: Stock and Watson (2002)

- Stock and Watson (2002b): $y^h_{t+h} = \alpha_h + \beta_h' \hat{F}_t + \sum_{j=0}^{p} \gamma_{hj} z_{t-j} + \varepsilon_{t+h}$ (direct forecast, one equation per h)
  - y^h_{t+h}: the target over the next h months, annualised; $\hat{F}_t$: the estimated factors; z_t: the target's own current growth; α_h, β_h, γ_{hj}: coefficients estimated by OLS for each h
  - real series: $y^h_{t+h} = \frac{1200}{h} \ln(Y_{t+h}/Y_t)$; prices: $\frac{1200}{h} \ln(P_{t+h}/P_t) - 1200 \ln(P_t/P_{t-1})$ (Y_t, P_t: levels)
  - DI: factors only, $k \in \{1, \ldots, 12\}$ by BIC; DI-AR, Lag: factors and $p \in \{0, \ldots, 5\}$ lags by BIC; benchmark: AR with BIC lags
- Recursive: factors re-estimated at each origin by EM on the data available then; origins from 1970:1
  - evaluation 1970–1998 (the paper) and 1999–2019 (extension), by forecast origin
- Targets: industrial production, payroll employment, CPI inflation; $h = 1, 6, 12$ months

Diffusion-index forecasts against an AR



- MSE relative to the AR forecast with BIC lags (below 1 = better)

ATS_ch5_factors

Interpreting the diffusion indexes

- 1970–1998: large gains for real activity at $h = 12$: IP 0.58, employment 0.62; for inflation only with own lags (DI-AR, Lag 0.68)
- 1999–2019: the gains vanish (IP 1.07, employment 1.09, inflation 1.35): the Great Moderation made activity less predictable from the panel
- BIC selects a median of 7 factors (10%–90%: 5–12) for DI-AR, Lag at $h = 12$
- Bayesian shrinkage and principal components give highly correlated forecasts in large panels De Mol, Giannone and Reichlin (2008): two ways to use the same information

FAVAR: Bernanke, Boivin and Eliasz (2005)



Ben Bernanke, 2008

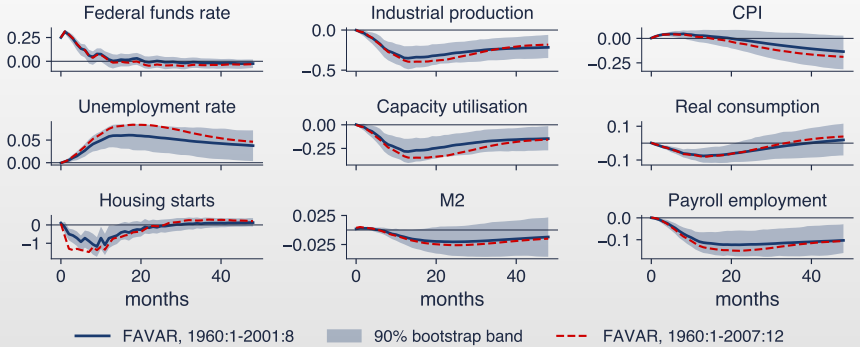
Photo: Federal Reserve (2008); public domain; Wikimedia Commons

- Bernanke, Boivin and Eliasz (2005)
  - VAR: $\binom{F_t}{Y_t} = \Phi(L)\binom{F_{t-\mathbf{1}}}{Y_{t-\mathbf{1}}} + v_t$; panel: $X_t = \Lambda^f F_t + \Lambda^y Y_t + e_t$
  - Y_t: the funds rate; F_t: K unobserved factors; X_t: the large panel
  - Λ^f, Λ^y: loadings; $\Phi(L)$: a lag polynomial; v_t, e_t: errors
- Why it helps
  - the VAR sees the information of 100 series through K factors
  - responses are available for every series in X_t

FAVAR: estimation and our replication

- Two-step estimation (Section III of the paper)
  - principal components $\hat{C}_t$ of X_t; slow factors from slow-moving series
  - remove Y_t: $\hat{F}_t = \hat{C}_t - \hat{b}_Y Y_t$; $\hat{b}_Y$: the coefficient of Y_t in a regression of $\hat{C}_t$ on the slow factors and Y_t
  - Cholesky identification with Y_t last; a 25 bp shock
- Specification
  - paper: $K = 3$, VAR(13), 1960:1–2001:8
  - ours: $T = 500$, $N = 102$ balanced series, of which 76 slow
  - prices and money enter as monthly growth rates; responses are cumulated to levels

FAVAR responses to a 25 bp monetary policy shock



- Levels in percent (rates and unemployment in pp); 90% residual-bootstrap bands, factors treated as data; dashed: sample to 2007:12

ATS_ch5_favar

Interpreting the FAVAR

- Real activity falls with a lag: IP trough -0.35% after 13 months, capacity utilisation -0.28 pp, payrolls -0.12%, unemployment up by 0.061 pp
- Housing starts react first and most (-1.18% after 9 months): the interest-sensitive sector
- The price puzzle is small (at most 0.046%, month 8); CPI ends at -0.14% after 48 months (band $[-0.32, 0.03]$); to 2007: -0.19%
- Seminar 5, B4: with $K = 1$ or $K = 5$ the puzzle returns; the choice of K is an identifying assumption, not a detail

The dynamic factor model (1/2)

- Measurement equation: $x_t = \Lambda f_t + e_t$, $e_t \sim N(0, \Psi)$
  - x_t: the N observed series; f_t: the r factors; Λ ($N \times r$): the loadings
  - e_t: idiosyncratic errors with diagonal covariance Ψ
- Transition equation: $f_t = A_1 f_{t-1} + \cdots + A_q f_{t-q} + u_t$, $u_t \sim N(0, Q)$
  - A_j: the factor VAR(q) matrices; Q: the covariance of the factor shocks u_t
- A state-space model (TSA, Chapter 10; Chapter 6)
  - state $s_t = (f_t', \ldots, f_{t-m}')'$, with $m \geq q - 1$ lags
  - estimated by the Kalman filter and smoother

The dynamic factor model (2/2)

- **Two-step estimation** Doz, Giannone and Reichlin (2011)
  - PCA for $\hat{f}_t$; OLS for Λ and Ψ; a VAR on $\hat{f}_t$ for A and Q
  - then the Kalman smoother re-estimates f_t
  - consistent for large N and T even though Ψ diagonal is misspecified
- **Quasi-maximum likelihood by EM** Doz, Giannone and Reichlin (2012), Bańbura and Modugno (2014)
  - E-step: the Kalman smoother; M-step: regressions on the smoothed moments
  - handles any pattern of missing data and idiosyncratic AR(1) terms

Missing data in the Kalman filter

- At time t, keep only the observed rows: $x_t^o = W_t x_t$, $\Lambda_t = W_t \Lambda$, $\Psi_t = W_t \Psi W_t'$ with W_t a selection matrix
  - if nothing is observed, the update step is skipped and the prediction is carried forward
- **Ragged edge**: at the end of the sample each series stops at a different month (publication lags); the filter uses whatever is there
  - the same mechanism handles series that start late (services confidence from 2002) and quarterly series observed every third month
- This is why nowcasting is a filtering problem, and why Chapter 6 develops the filter in full

The nowcasting problem



Lucrezia Reichlin, 2013

Photo: Niccolò Caranti (2013); CC BY-SA 3.0; Wikimedia Commons

- **Nowcasting**: estimating the present and the recent past of a low-frequency variable (GDP) from timelier high-frequency data Giannone, Reichlin and Small (2008), Bańbura, Giannone, Modugno and Reichlin (2013)
  - backcast (last quarter, not yet published), nowcast (this quarter), forecast (next quarter)
- Three difficulties: mixed frequencies, many indicators, asynchronous releases (the ragged edge)
  - surveys are out at the end of the month; industrial production more than a month later; GDP weeks after the quarter
- Tools: bridge equations Baffigi, Golinelli and Parigi (2004), MIDAS Ghysels, Santa-Clara and Valkanov (2006), Ghysels, Sinko and Valkanov (2007), mixed-frequency DFM Mariano and Murasawa (2003), Bańbura and Modugno (2014) and mixed-frequency VARs Schorfheide and Song (2015)

Bridge equations and MIDAS

- **Bridge**: forecast the missing months of each indicator (AR), aggregate to the quarter, regress GDP on the quarterly aggregates Baffigi, Golinelli and Parigi (2004)
  - simple and transparent; the error of the monthly AR forecasts passes into the nowcast
- **MIDAS** Ghysels, Santa-Clara and Valkanov (2006): $y_\tau = \beta_0 + \beta_1 \sum_{j=0}^{K-1} w_j(\theta) x_{m_\tau - j} + \rho y_{\tau - s} + \varepsilon_\tau$, with the latest monthly value x_{m_τ}
  - τ: the quarter; m_τ: its latest available month; x: a monthly indicator with K lags; $w_j(\theta)$: weights summing to 1; $y_{\tau-s}$: lagged GDP growth
  - exponential Almon: $w_j(\theta) = e^{\theta_1 j + \theta_2 j^2} / \sum_k e^{\theta_1 k + \theta_2 k^2}$; two parameters for K lags, estimated by NLS
  - MIDAS with leads Andreou, Ghysels and Kourtellos (2013), Clements and Galvão (2008): one regression per information set; U-MIDAS leaves w_j unrestricted when K is small Foroni, Marcellino and Schumacher (2015)
- Neither uses the joint dynamics of the indicators; the DFM does

Linking months and quarters: Mariano and Murasawa (2003)

- Let Y_t^M be a latent monthly level and the quarterly GDP the 3-month average, observed in the third month
  - geometric-mean approximation: $\ln Y_t^Q \approx \frac{1}{3}(\ln Y_t^M + \ln Y_{t-1}^M + \ln Y_{t-2}^M)$
  - quarterly growth $y_t^Q = \ln Y_t^Q - \ln Y_{t-3}^Q = \frac{1}{3}(y_t + 2y_{t-1} + 3y_{t-2} + 2y_{t-3} + y_{t-4})$, y_t = monthly growth
    Mariano and Murasawa (2003)
- In the DFM: $y_t^Q = \lambda_y' \sum_{j=0}^{4} w_j f_{t-j} + e_t^Q$, $w = (1, 2, 3, 2, 1)/3$: the state must hold five lags of f_t
  - λ_y: the loadings of GDP growth; w_j: the Mariano–Murasawa weights; e_t^Q: the idiosyncratic error of GDP
  - GDP is a monthly variable observed every third month: missing in months 1 and 2 of each quarter
- Seminar 5, A7: the weights by hand

Case study: nowcasting Romanian GDP

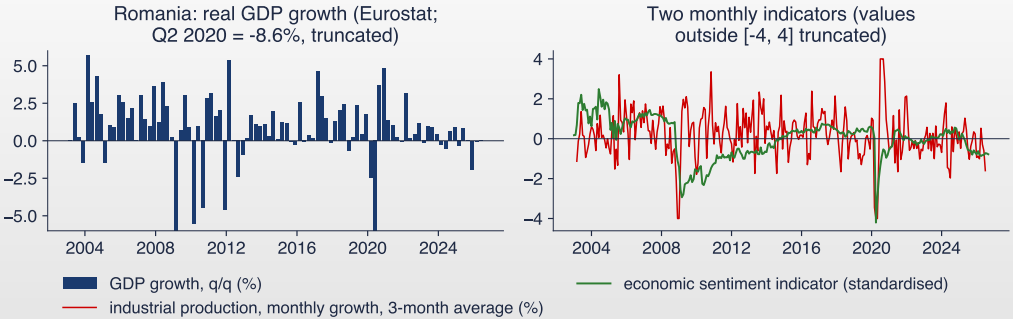


Institutul Național de Statistică, Bucharest

Photo: Dan Mihai Pitea (2009); CC BY-SA 3.0; Wikimedia Commons

- Target: q/q growth of real GDP (Eurostat), 94 quarters to 2026Q2; monthly panel: 11 series from 2003 (Eurostat)
  - hard data in monthly growth: IP, retail, construction, euro-area IP; change of unemployment; levels of six confidence balances
- Stylised release lags at the end of each month: surveys 0, unemployment 1, hard data 2 months; GDP 2 months after the quarter
  - current-vintage data: pseudo real time (no revisions)
- Models: AR(1), bridge (IP, retail, ESI), MIDAS (IP, ESI; $K = 6$), DFM with two factors (two-step and EM)

Romanian GDP and two monthly indicators



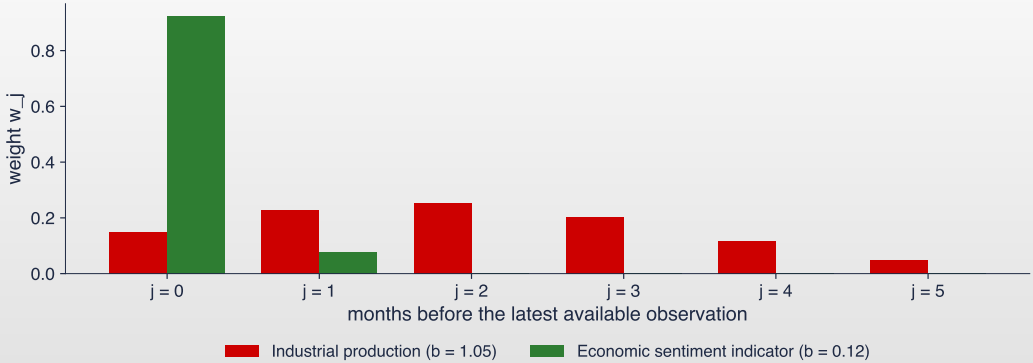
⊡ Quarterly GDP growth (2020Q2: −8.6%, truncated); IP growth (3-month average) and the standardised ESI

ATS_ch5_nowcast

Interpreting the Romanian data

- Quarterly growth is volatile: 2009–2012 swings of ± 5 pp and the 2020 collapse dominate the variance
- The ESI (3-month average) has correlation 0.44 with GDP growth before 2020: informative, far from perfect
- Latest published quarter: 2026Q2, growth 0.00%; the nowcast target is 2026Q3

MIDAS weights for Romania



⊡ ADL-MIDAS with exponential Almon weights at the end of the third month of the quarter, 2003–2026

ATS_ch5_nowcast

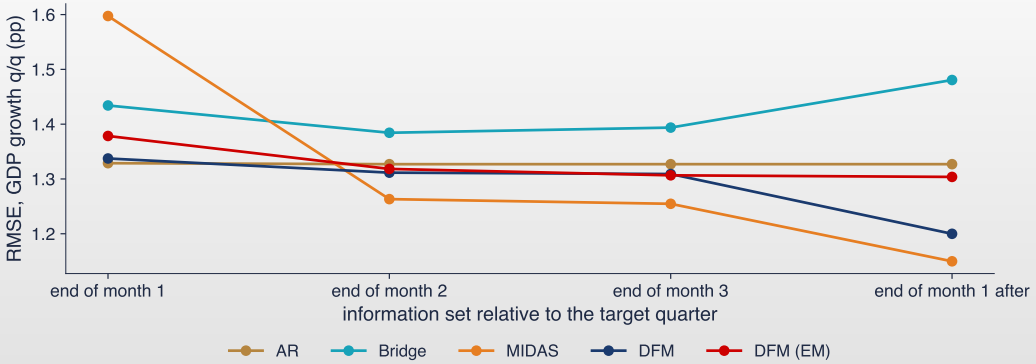
Interpreting the MIDAS weights

- IP: hump-shaped weights peaking at lag 2 (0.25): the months of the quarter itself matter, as the Mariano–Murasawa weights predict
- ESI: weight 0.92 on the latest month: a level indicator already summarises the past
- Lagged GDP growth has coefficient -0.28: the noisy q/q series mean-reverts

Pseudo-real-time evaluation

- For each quarter 2013Q1–2026Q2, nowcasts at the ends of months 1, 2, 3 of the quarter and of month 1 after it
  - all parameters re-estimated on the data available at that date; 2020Q2–Q3 excluded from the RMSE (also reported with them)
- The evaluation sample starts in 2013 because the 2009–2012 swings are larger than any model error; from 2010 the RMSEs rise to about 2 pp for every model
  - standard deviation of the target 2013–2026 (without 2020Q2–Q3): 1.33 pp
- Question: does the RMSE fall as the quarter unfolds, and by how much relative to an AR?

Nowcast accuracy by information set



- RMSE of the nowcasts of q/q GDP growth, 2013Q1–2026Q2 without 2020Q2–Q3

ATS_ch5_nowcast

Interpreting the nowcast accuracy

- AR: 1.33 pp throughout; DFM: 1.34 (month 1) to 1.20 (month 1 after); MIDAS: 1.60 to 1.15
- Bridge does not beat the AR (1.39); the EM estimate of the DFM is close to the AR (1.31 at month 3)
- Diebold–Mariano DFM against AR at month 3: $t = -0.30$, $p = 0.76$: the gain is not significant on 54 quarters
- With 2020: AR 1.90, DFM 1.30 at month 1 after: the monthly data catch the collapse, the AR cannot

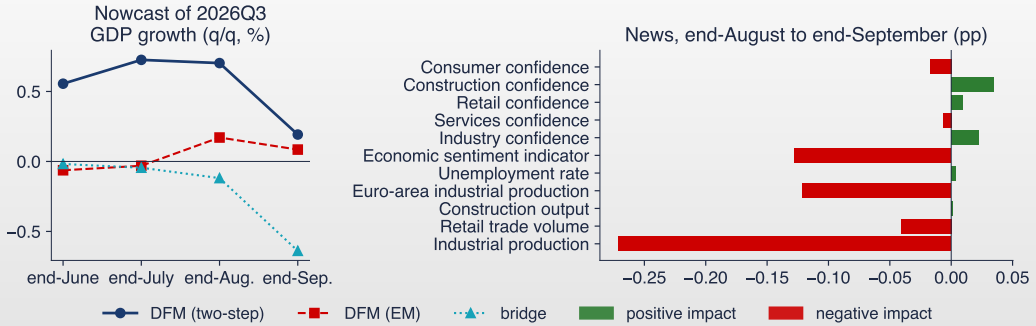
News and revisions: Bańbura and Modugno (2014) (1/2)

- The **news** of a release: $l_j = x_j - \mathbb{E}[x_j \mid \Omega_v]$, not x_j itself
  - $\Omega_v \subset \Omega_{v+1}$: two successive information sets (data vintages)
  - x_j, $j \in J_{v+1}$: the releases that are new in Ω_{v+1}
- With fixed parameters, the revision is a weighted sum of the news Bańbura and Modugno (2014)
  - $\mathbb{E}[y \mid \Omega_{v+1}] - \mathbb{E}[y \mid \Omega_v] = \mathbb{E}[yl'][ll']^{-1}l = \sum_j w_j l_j$
  - y: the target (GDP growth); l: the vector of news l_j; w_j: the weight of release j
- A large release that was expected moves nothing; a small surprise in a heavily weighted series moves a lot; correlated releases share their weight

News and revisions: Bańbura and Modugno (2014) (2/2)

- In the DFM: $l_j = \lambda_j'(f_{t_j} - \hat{f}_{t_j|v}) + e_j$, so $\mathbb{E}[ll'] = HP_{|v}H' + \Psi_J$ and $\mathbb{E}[yl'] = z_y' P_{|v} H'$
  - λ_j: the loadings of release j; t_j: its reference month; $\hat{f}_{t_j|v}$: the factor estimate given Ω_v; H: the stacked λ_j'; Ψ_J: their idiosyncratic variances; z_y: the loading of the target
  - $P_{|v}$: the covariance of the stacked state given Ω_v, from the Kalman filter

Nowcasts for 2026Q3 and the news decomposition of the revision



Left: nowcasts at the ends of June–September 2026; right: news decomposition of the two-step DFM nowcast between the August and September information sets

ATS_ch5_nowcast

Interpreting the news

- DFM nowcast of 2026Q3: 0.70% at the end of August, 0.19% at the end of September; the 11 new releases add up to -0.51 pp exactly
- The July IP release was -3.7 pp below the model's expectation; with weight 0.073 it cut the nowcast by -0.27 pp; ESI -0.13, euro-area IP -0.12
- Other models at the end of September: EM DFM 0.08%, bridge -0.64%, AR 0.76%: the first official estimate for 2026Q3 will judge them
- The news decomposition turns a number into a story that a central bank can check release by release

An open question

- Can a nowcasting model beat a simple AR for Romanian GDP growth in a statistically significant way, and which releases carry the information?
  - formal: H_0: equal MSE of the DFM and AR nowcasts at the end of month 3 (Diebold–Mariano, Chapter 1), pre-registered sample, models and lags
  - falsified by a rejection on quarters the model never saw, with real-time vintages of GDP
- Why it matters: fiscal and monetary decisions in 2025–2026 were taken with GDP known only weeks after each quarter
  - literature to start from: Giannone, Reichlin and Small (2008), Bańbura and Modugno (2014), Bańbura, Giannone, Modugno and Reichlin (2013), Giannone, Lenza and Primiceri (2021), Schorfheide and Song (2015)

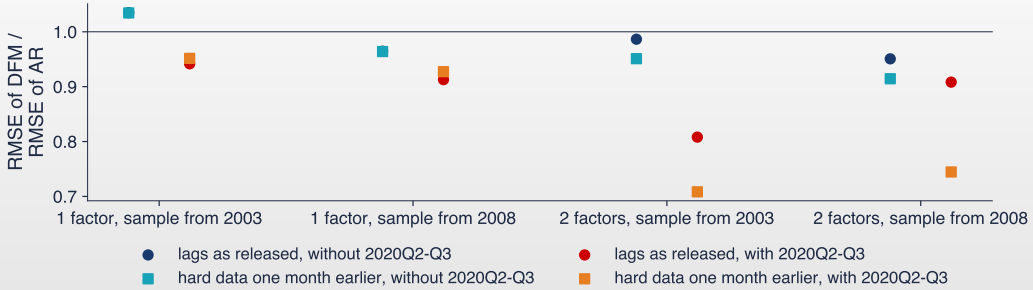
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed papers that nowcast GDP of Central and Eastern European countries with factor models or MIDAS; give DOIs.` Then check every DOI on Crossref
  - **hypothesis**: `Which Romanian monthly releases should carry news about GDP, and in which month of the quarter?`
  - **code and replication**: ask for a Kalman filter with missing data, then reproduce a known number first (the news sum of this lecture)
  - **robustness and critique**: `Act as a hostile referee of a nowcasting paper: list the ways the pseudo-real-time design could flatter the model.`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- The information set at each date contains only data published by then; current-vintage data overstate accuracy
- Hyperparameters, factors and lags are chosen inside each estimation window, never on the evaluation sample
- The evaluation period and the treatment of 2020 are fixed before the results; all variants are reported
- A claim of "better nowcasts" is backed by a test, not by a lower RMSE alone

Mini-case: sensitivity of the nowcast ranking to the specification



- RMSE of the DFM nowcast relative to the AR at the end of month 3, 2013–2026, across factors, sample start, release lags and the treatment of 2020
- 16 variants range from 0.71 to 1.03; the DFM wins in 14 of them: an AI summary that reports the best variant as "the" result is wrong

ATS_ch5_nowcast

Project idea

- **A real-time nowcasting model for Romanian GDP**: replicate first, then extend
  - replicate: the news decomposition of this lecture (-0.51 pp between August and September 2026) and the BGR table for 1971–2003
  - extend: real release dates (INS calendar), GDP vintages saved from now on, blocks of factors for hard and soft data, a mixed-frequency BVAR, density nowcasts
  - pre-register: target, quarters, models, information sets, the treatment of 2020 and the Diebold–Mariano test
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- Many series and few observations: shrink (priors) or compress (factors)
- The Minnesota prior in conjugate form gives closed-form posteriors, exact draws and a marginal likelihood that chooses the shrinkage
- Bigger systems need tighter priors; then they forecast better and give more plausible responses
- Principal components estimate the factor space; Bai–Ng chooses its dimension; FAVAR and DFM put it to work
- Nowcasting is Kalman filtering with a ragged edge; every revision is a sum of news

Self-assessment

Questions

- What happens to the Minnesota posterior mean when $\lambda \to 0$?
- Why does the natural conjugate prior force $\vartheta = 1$?
- Which prior allows cointegration: sum-of-coefficients or initial observation?
- Why can PCA identify only the factor space?
- Why is a large but expected release not news?

Next:

- State space models and Bayesian filtering
- The Kalman filter and smoother in full; EM, particle filters, stochastic volatility

Appendix: the natural conjugate posterior

- Likelihood: $p(Y \mid B, \Sigma) \propto |\Sigma|^{-T/2} \exp\{-\frac{1}{2}\mathrm{tr}[\Sigma^{-1}(Y - XB)'(Y - XB)]\}$
- Prior: $|\Sigma|^{-k/2} \exp\{-\frac{1}{2}\mathrm{tr}[\Sigma^{-1}(B - B_0)'\Omega_0^{-1}(B - B_0)]\} \times |\Sigma|^{-(d+n+1)/2} \exp\{-\frac{1}{2}\mathrm{tr}(\Sigma^{-1}\Psi)\}$
- Complete the square in B:
  $(Y - XB)'(Y - XB) + (B - B_0)'\Omega_0^{-1}(B - B_0) = (B - \bar{B})'\bar{\Omega}^{-1}(B - \bar{B}) + \hat{U}'\hat{U} + (\bar{B} - B_0)'\Omega_0^{-1}(\bar{B} - B_0)$
- Hence $B \mid \Sigma, Y \sim MN(\bar{B}, \bar{\Omega}, \Sigma)$ and $\Sigma \mid Y \sim IW(\bar{\Psi}, T + d)$: the same families as the prior
- $MN(\bar{B}, \bar{\Omega}, \Sigma)$: the matrix normal distribution, i.e. $\mathrm{vec}(B) \sim N(\mathrm{vec}(\bar{B}), \Sigma \otimes \bar{\Omega})$

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Appendix: the marginal likelihood in closed form

- Integrating B and Σ out (the online appendix of GLP):
  $p(Y) = \pi^{-nT/2} \frac{\Gamma_n(\frac{T+d}{2})}{\Gamma_n(\frac{d}{2})} |\Omega_0|^{-\frac{n}{2}} |X'X + \Omega_0^{-1}|^{-\frac{n}{2}} |\Psi|^{\frac{d}{2}} |\bar{\Psi}|^{-\frac{T+d}{2}}$
- $\Gamma_n(a) = \pi^{n(n-1)/4} \prod_{j=1}^{n} \Gamma(a + (1-j)/2)$: the multivariate gamma function
- $|\Omega_0|\,|X'X + \Omega_0^{-1}| = |I + \Omega_0 X'X|$: the complexity penalty, growing with λ
- $|\bar{\Psi}|$: the fit term, falling with λ; the optimum balances the two
- With dummies: $\ln p(Y \mid \gamma) = \ln p(Y, Y_d \mid \gamma) - \ln p(Y_d \mid \gamma)$, each by the formula above

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